

Robert Serber *(left)* and Robert Oppenheimer *(right)* discussing physics, ca. 1942. [Courtesy D.S. Information Agency.]

Oppenheimer and Serber quickly realized that, if the Sun has a neutron core at its center, and if the core's mass is a large fraction of the Sun's mass, then the core's intense gravity will hold the Sun's outer layers in a tight grip, making the Sun's circumference far smaller than it actually is. Thus, Landau's neutron-core idea could work only if neutron cores can be far less massive than the Sun.

"How *small* can the mass of a neutron core be?" Oppenheimer and Serber were thus driven to ask themselves. "What is the *minimum* possible mass for a neutron core?" Notice that this is the *opposite* question to the one that is crucial for the existence of black holes; to learn whether black holes can form, one needs to know the *maximum* possible mass for a neutron star (Figure 5.3 above). Oppenheimer as yet had no inkling of the importance of